Math 110 – Homework 4 Answers

Math 110 – Fall 2026

Homework 4

Isomorphisms, inverses, products, and quotient spaces

Suggested completion date: Friday, September 25

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(a) Verify by direct multiplication that

$$A \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I.$$

(b) Define $\det A = ad - bc$. Prove that if $\det A \neq 0$, then A is invertible and derive the usual formula for A^{-1}.

(c) Prove conversely, without using any general determinant theory, that if $\det A = 0$, then A is not invertible.

Problem 2. Use the result of Problem 1 to decide which of the following matrices are invertible, and find the inverse whenever it exists:

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix} \text{ over } \mathbb{C}, \qquad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} \text{ over } \mathbb{F}_5.$$

Problem 3. (LADR §3D, Exercise 3.) Let $T : V \to V$ be linear and suppose V is finite-dimensional. Prove that the following are equivalent:

(a) T is an isomorphism;

(b) for every basis $v_1, \ldots, v_n$ of V, the list $Tv_1, \ldots, Tv_n$ is a basis of V.

Problem 4. Suppose B and C are matrices for which BC is defined.

(a) Prove

$$\operatorname{rank}(BC) \leq \min\{\operatorname{rank} B, \operatorname{rank} C\}.$$

(b) Give an example in which the inequality is strict even though both B and C have positive rank.

Problem 5. Let V and W be finite-dimensional vector spaces. Prove

$$\dim(V \times W) = \dim V + \dim W$$

by constructing a basis of $V \times W$ from bases of V and W.

Problem 6. Let U be a subspace of V and let $v, v' \in V$.

(a) Prove that the affine set $v + U$ is a subspace of V if and only if $v \in U$; in that case $v + U = U$.

(b) Prove that $v + U = v' + U$ if and only if $v - v' \in U$.

Problem 7 (optional). Suppose U is a subspace of a finite-dimensional vector space V, and $S : U \to V$ is injective. Prove that S can be extended to an invertible operator $T : V \to V$.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
